



Iowa City Robotics

EV3 Robot Challenge (Prebuilt)

Summary

Students use a pre-built LEGO robot to complete simple races and challenges. They learn how robots follow instructions and how small changes can change what a robot does.

Description

Students work in small groups to control a LEGO EV3 robot. The robot is already built and programmed with simple buttons or programs. Students can race robots, drive through obstacle courses, or complete small challenges. This activity helps students explore robotics, teamwork, and problem solving.

Materials

Material	Quantity
Pre-built LEGO EV3 robot	1 per group
EV3 program or remote control	1 per robot
Masking tape (to mark start/finish lines)	1 roll per classroom
Small objects or cones (for obstacles)	Several per classroom
Measuring tape or ruler	1 per classroom

Directions

- Divide students into small groups (2–3 students)
- Show students the robot and explain the basic parts
 - Wheels
 - Motors
 - Buttons or program that controls it
- Demonstrate how to start the robot
- Create a start line and finish line on the floor using tape



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- Place each robot at the start line
- Start the robots and watch them race to the finish line
- Let students take turns starting or driving the robot
- After the race, let students try again and see if they can improve their robot's path

Challenge Ideas

- **Robot Race** See which robot reaches the finish line first
- **Obstacle Course** Add cones, books, or blocks for robots to drive around
- **Target Stop** Try to stop the robot as close as possible to a line or object
- **Robot Delivery** Place a small object on the robot and try to deliver it to a finish line

Activity Extension

- Change the distance of the race and see how it affects the robot
- Let students guess where the robot will stop before starting it
- Try different programs that change how long or fast the robot moves
- Add simple sensor challenges (stop when it sees a color or object)

What is Happening?

- The robot follows instructions from its program
- Motors spin the wheels so the robot can move
- Changing the program changes how far or how fast the robot moves
- Engineers test robots many times to improve how they work